

Question	Working	Answer	Mark	Notes
22	$\cos PBQ = \frac{10^2 + 10^2 - x^2(2 - \sqrt{3})}{200}$ $= \frac{200 - x^2(2 - \sqrt{3})}{200}$	Proof	B1 M1 M1 M1 A1	(indep) for stating $\cos 30 = \frac{\sqrt{3}}{2}$ for $PQ^2 = 10^2 + 10^2 - 2 \times 10 \times 10 \times \cos PBQ$ or $AC^2 = x^2 + x^2 - 2 \times x \times x \times \cos 30 (=x^2(2-\sqrt{3}))$ oe for $\cos PBQ = \frac{10^2 + 10^2 - PQ^2}{2 \times 10 \times 10}$ (implies previous M1) for $\cos PBQ = \frac{10^2 + 10^2 - (x^2 + x^2 - 2 \times x \times x \times \cos 30)}{2 \times 10 \times 10}$ conclusion of proof with all working seen